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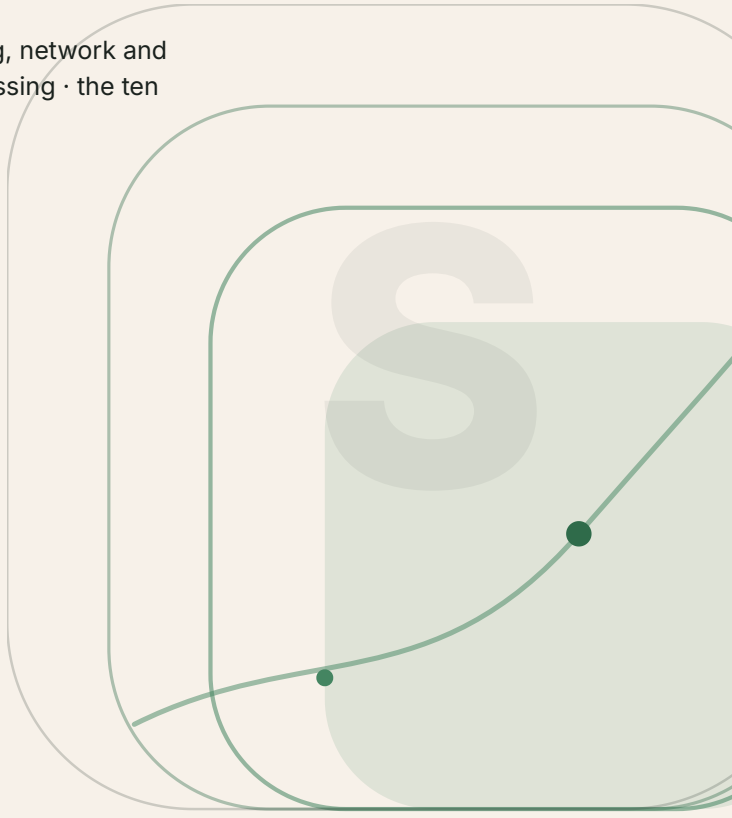
S

SITE PACK S

# Solar plant with grid connection

Solar and grid, with an optional battery. Everything for one kind of site in one pack: what to buy, the survey, the diagrams, install day step by step, the platform steps, the commissioning tests and the handover sheet.

| | |
|---|---|
| FOR | The project manager at this kind of site, the electrician, and the customer's owner at handover |
| PRINT | The whole pack, for the site in front of you. Tick the boxes as you go |
| COVERS | Bill of materials · survey · system, wiring, network and enclosure diagrams · install day · addressing · the ten setup steps · commissioning · handover |



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Greecon Sh.p.k · Durana Tech Park · info@greecon.earth

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SITE PACK S

# Solar plant with grid connection

*Solar and grid, with an optional battery.* A complete pack for one kind of site, in the order of the work: what to buy, the survey, the drawings, the office, install day step by step, the platform's setup list, the commissioning tests and the handover. Print all of it and tick the boxes as you go. The electrician does everything on the mains side; you do the low voltage and the software; Greecon only reads.

## At a glance

**WHO IT SUITS**
A solar plant with a grid connection, with or without a battery. It works with any inverter brand, because it measures at the AC cable with its own meter and not through the inverter maker's portal.

**WHAT IT MEASURES**
Solar production, grid import and export, and, if wanted, the site's own consumption (a load meter) and a battery's charge and power.

**PLANNING ESTIMATE**
About EUR 600 to 1,200.

**TIME**
Survey about 1 hour. Install day 4 to 6 hours with the electrician. Commissioning about 1 hour. Handover 30 minutes.

**TEMPLATE IN THE PLATFORM**
**Solar plant with grid connection**

**STANDARD SETTINGS**
Unit numbers: grid meter 1, solar meter 2, optional load meter 4. 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode.

**The order of the work:** survey, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

**Six rules for the whole job.**

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable between meters, the 4 to 20 mA signal wires, and the 12 or 24 V supply from a plug-in supply.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

## How to read this pack

| Mark | What it means |
|---|---|
| ELECTRICIAN | The electrician. Everything inside a distribution board, every current clamp and voltage tap, and the pump's run contact. |
| PROJECT MANAGER | The project manager: low voltage and software. |
| CUSTOMER | The customer: access, people, and starting or stopping their own equipment. |
| ☐ | A box to tick when it is done. Print the pack and tick as you go. |
| **Do**, **Check**, **Stop**, **Tip**, **Say** | The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer. |
| Figure S-1 and so on | Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual. |

# What it measures

The template **Solar plant with grid connection** makes up to four devices. Delete what the site does not have (see the project manager's steps below). What each gives:

| Reading in Greecon | Comes from | Unit | Standard alert limits |
|---|---|---|---|
| **Grid import** and **Grid export** | Grid meter: an Eastron SDM630 where the supply enters, unit 1. One meter gives both: positive power is energy coming in, negative is energy sent out | kW | None |
| **Solar production** | Solar meter: an Eastron SDM630 on the inverter's AC output, unit 2. It reads the size of the power, so it works with any inverter brand | kW | None |
| **Energy consumption** (optional) | Load meter (optional): an Eastron SDM630 on the feed to the loads, unit 4, if the owner wants the site's own consumption measured | kW | None |
| **Battery charge** (optional) | Battery system (optional): the battery's own Modbus, mapped by hand from the maker's register table | % | Low below 25 % |
| **Battery power** (optional) | The same device. Positive while charging | kW | None |

If the maker of the battery will not publish its Modbus register table, leave the battery out: do not guess registers. Solar surplus has no standard limit: the platform says nothing about it rather than guess.

# What to buy and bring

| | Item | Specification | Qty | Planning estimate (EUR) |
|---|---|---|---|---|
| ☐ | **Box** | Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input | 1 | 180 to 300 |
| ☐ | **Converter** | RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode | 1 | 30 to 60 (basic) or 250 and up (industrial) |
| ☐ | **Three-phase meter** | Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *Grid and solar; a third on the site's own consumption if the owner wants it measured.* | 2, or 3 | 90 to 180 each |
| ☐ | **24 V DIN power supply** | 30 to 60 W | 1 | 25 to 50 |
| ☐ | **UPS** | A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS | 1 | 60 to 150 |
| ☐ | **Enclosure and fittings** | IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands | 1 | 80 to 150 |
| ☐ | **Cable and labels** | Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels | 1 lot | 40 to 80 |
| ☐ | **Cellular router** *(optional)* | 4G, DIN rail, data SIM. *Only without usable internet; not in the hardware guide's list for this kit, see the network rules.* | 0 or 1 | 150 to 200 |

**About EUR 600 to 1,200.** For a battery add nothing but the maker's Modbus register table. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in the Hardware Reference (PDF 03).

## The project manager brings

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to talk to a meter alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter
- ☐ Screwdrivers (slotted and Phillips, small and medium)
- ☐ Wire strippers and a ferrule crimper
- ☐ A network cable tester
- ☐ Cable ties and labels
- ☐ A head torch
- ☐ Spare fuses and terminals
- ☐ A phone for photographs

**You do not need a tool for work inside a live board: that is the electrician's.** The spares kit (one for each project manager) goes in the car too.

# Survey

About one hour, with the electrician if the plan needs one. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

## Ask, in this order

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each of them also needs a login of their own, so that they can acknowledge an alert, and a login needs an email address.
2. What went wrong in the last twelve months that cost money? Write it in their words. It is the reason they buy.
3. The electricity price per kWh, **what is paid for energy sent to the grid, if anything,** and the last twelve bills. The platform shows what solar kept on site at their price; without a price it shows kWh only.
4. What brand is the inverter, and where is its AC output? **Is the plant allowed to send energy to the grid?** If not, tick *This site cannot send energy to the grid* in the prices.
5. Where does the supply enter the building? Is it three-phase?
6. Is there a battery? Which maker and model, and does the maker publish the Modbus register table? Does it speak Modbus on the site's network, or on its own RS-485 port?
7. Does the owner want the site's own consumption measured (a third meter)?
8. Is there a network socket where the box will sit, or cellular signal? (Look at your phone there.)
9. Is there a power socket for the box, and is it on a circuit that stays on?
10. Who is the customer's electrician, and can they be there on install day?

## Look at, and photograph

- The supply entry and the main board (closed): where a meter could be fitted with the clamps around the incoming phases.
- The inverter and its AC output cable.
- The battery and its communication port, if there is one. Do not open it.
- The place for the box: dry, 30 cm free, no heat, within reach of a socket and a network cable.
- The route for the data cable from the board to the box (under 100 metres; note any crossing of power cable).

CHECK

**Go** if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

**No-go, politely, and write why,** if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

# Survey form

KIT S · SOLAR PLANT WITH GRID CONNECTION

| Field | Answer |
|---|---|
| Customer, site name and address, date | |
| Contact: name, mobile, email (first, second, third to be told) | |
| What went wrong last year (their words) | |
| Prices from the bills: electricity per kWh; paid for energy sent to the grid | |
| Twelve electricity bills collected | ☐ Yes ☐ Later |
| Allowed to send energy to the grid? | ☐ Yes ☐ No (tick the box in the prices) |
| Supply: single or three phase; main breaker size (A) | |
| Inverter: brand, size (kW), where its AC output cable is | |
| Battery: maker and model; register table available; connection (network or RS-485) | ☐ No battery |
| Meters needed: grid, solar, load (and where) | |
| Network: customer's cable or cellular; router brand; who can give an address | |
| Power for the box: socket location | |
| Electrician: name, mobile, day | |
| Photographs taken (list) | |
| Go / no-go, and why | ☐ Go ☐ No-go |

# The drawings

Every drawing is for kit S and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers. Mains work is inside the dashed red box and is the electrician's.

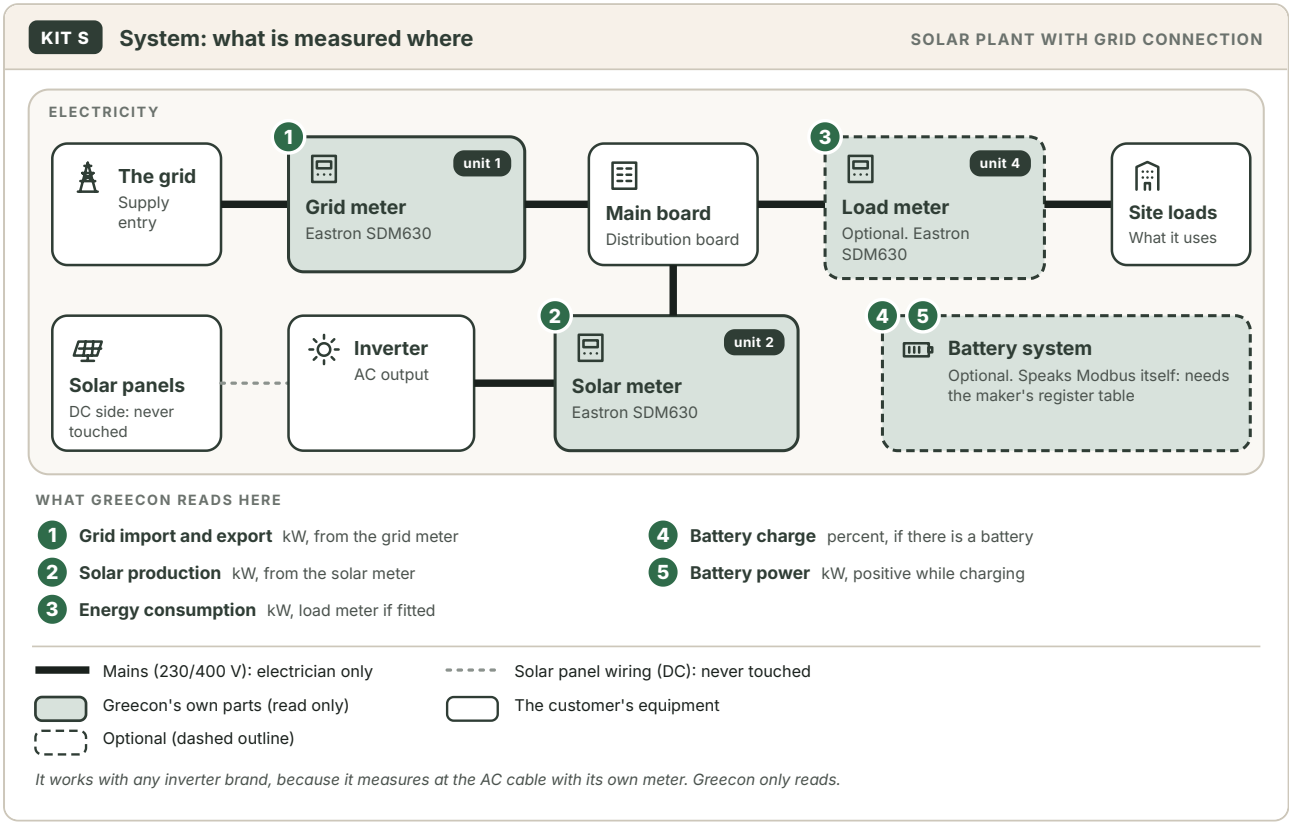



**Figure S-1.** Kit S, solar plant with grid connection. System: what is measured where. The dashed boxes are optional.

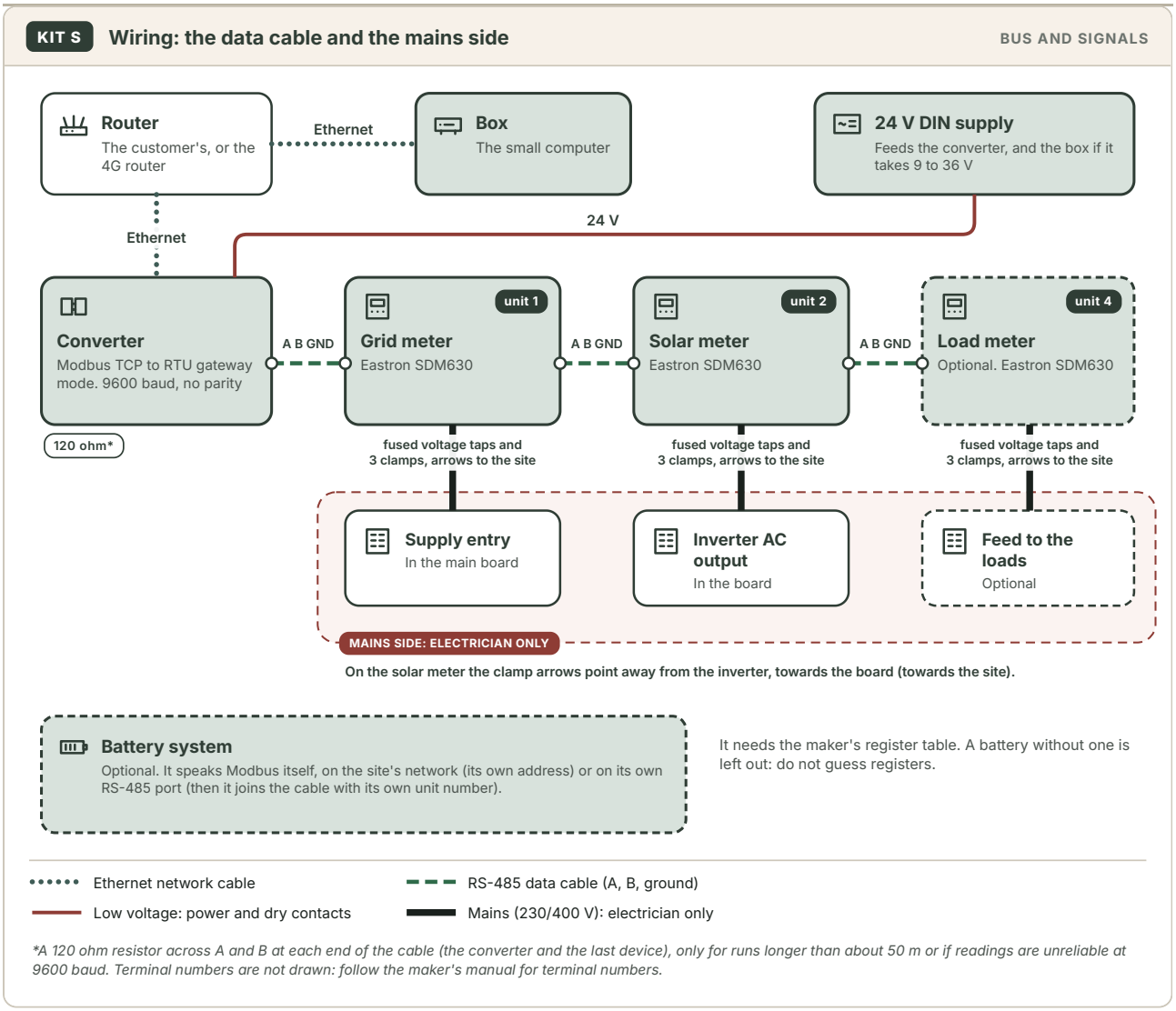



**Figure S-2.** Kit S. Wiring: one RS-485 cable from the converter through the grid meter, the solar meter and, if fitted, the load meter. Mains work is inside the dashed box.

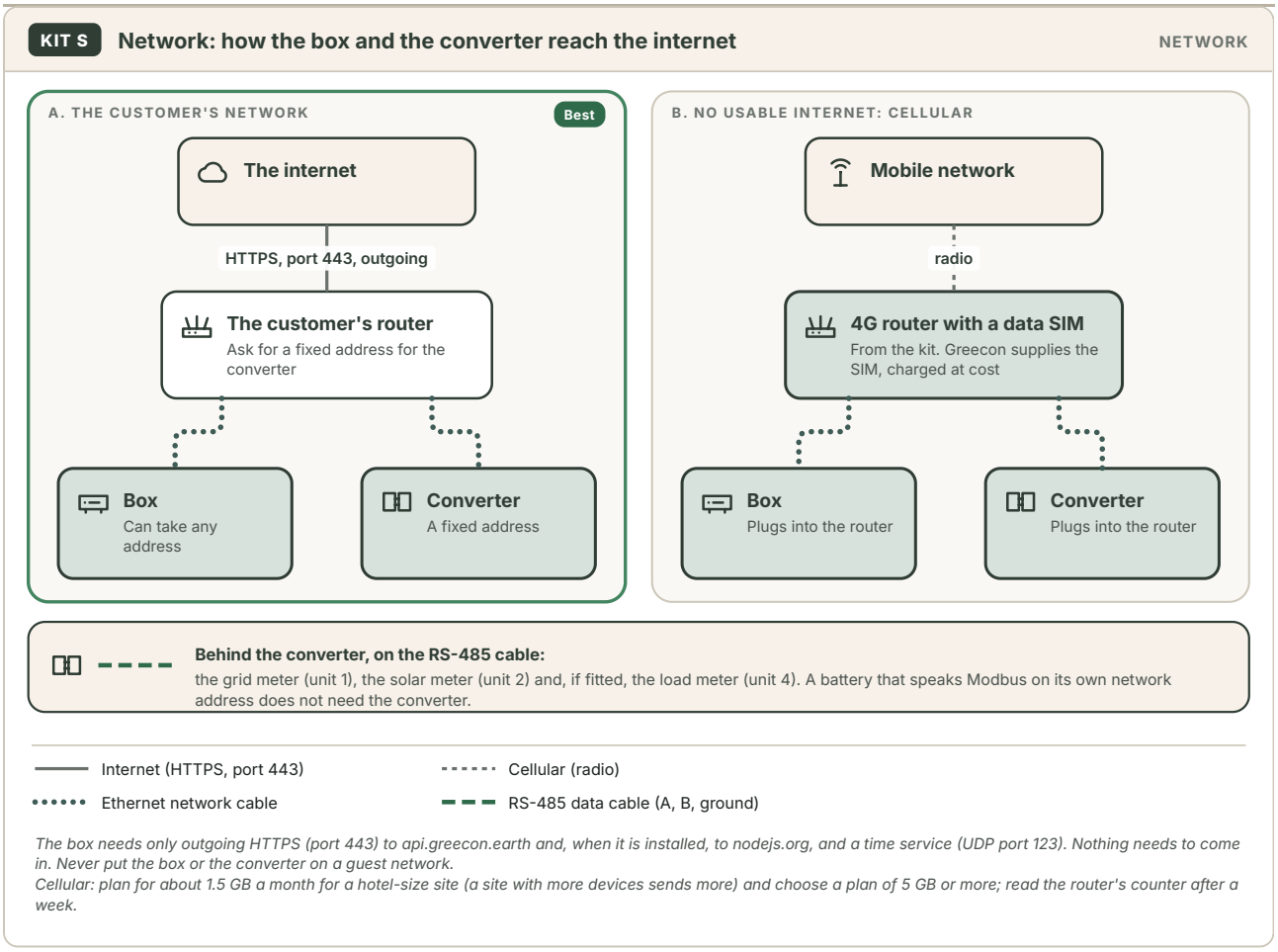



**Figure S-3.** Kit S. Network: the box and the converter on the customer's router (best) or on a 4G router.

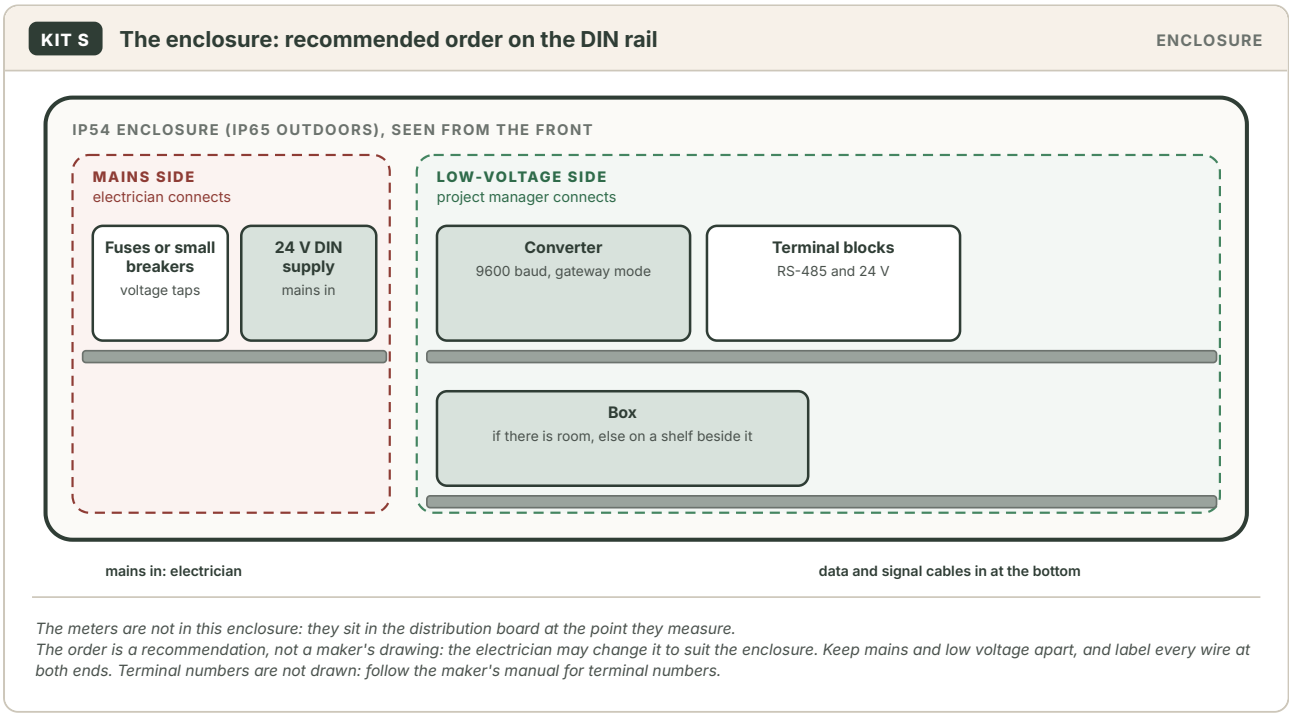



**Figure S-4.** Kit S. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

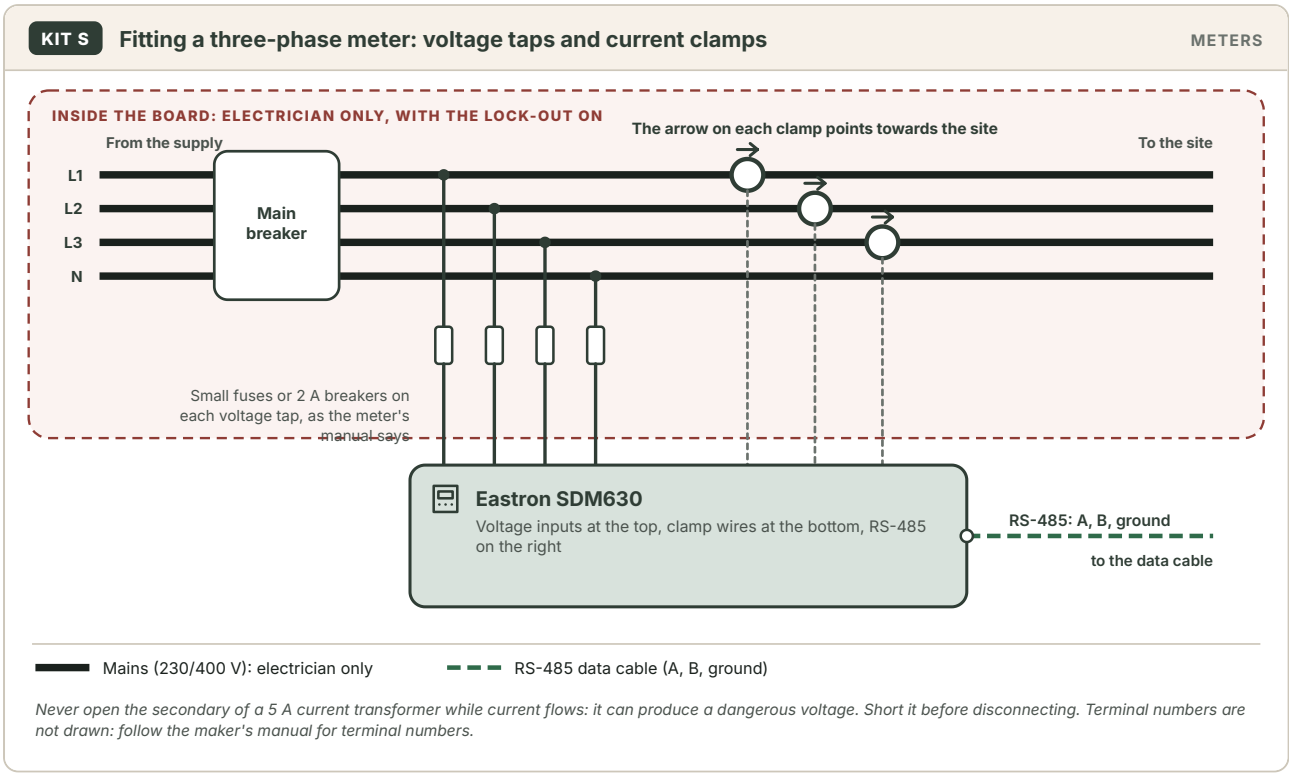



**Figure S-5.** Kit S. Fitting the grid meter and the solar meter (and the load meter): fused voltage taps and three clamps each, arrows towards the site.

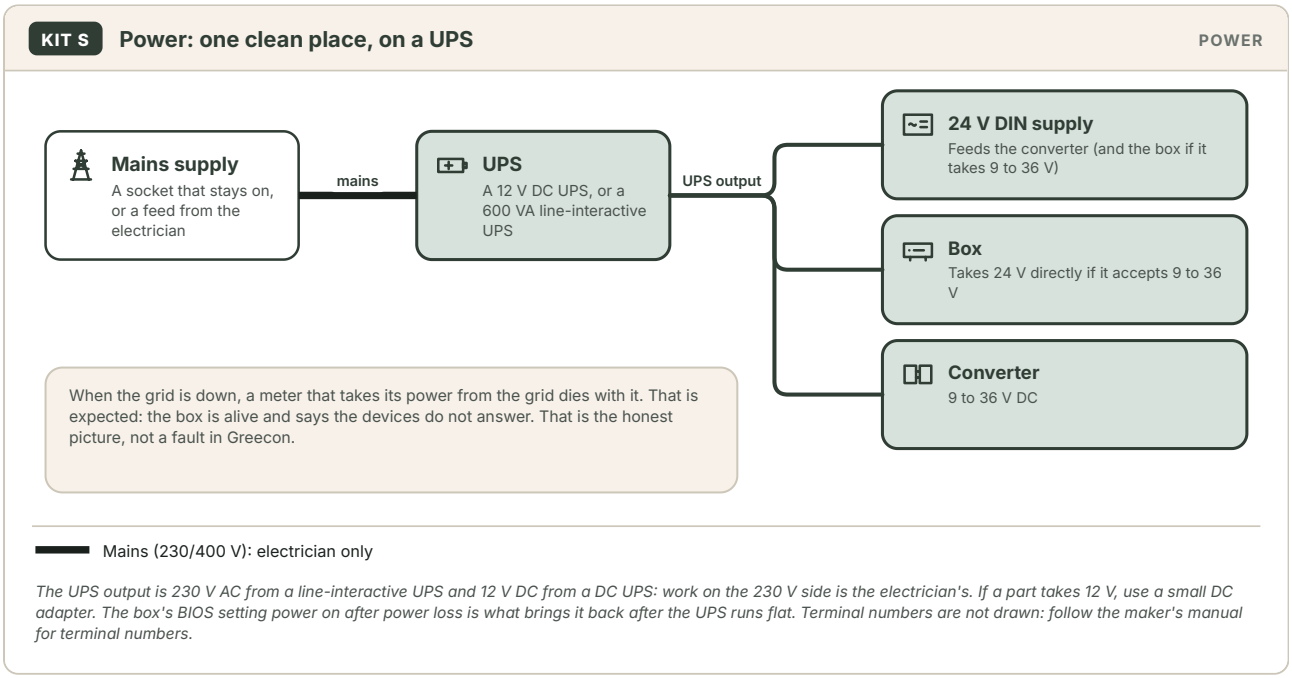



**Figure S-6.** Kit S. Power: the UPS feeds the box, the converter and the 24 V supply.

# At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box. PROJECT MANAGER

## In Greecon

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.
- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.
- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.
- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Solar plant with grid connection**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.
- [ ] **Get the install line:** you land on **Set up** and the site's name, with `0 of N steps done`. Press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> STOP
>
> **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

## The box on your desk

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in the Hardware Reference (PDF 03), section 4.3.
- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).
- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`

- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit and the printed pack.

TIP

**If the line stops with `STOP:`** it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

**The first time, rehearse the whole job on a desk** with a laptop standing in for the meters: the Project Manager Field Manual (PDF 02), section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

## Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

| | # | Who | Do this |
|---|---|---|---|
| ☐ | 1 | ELECTRICIAN, PROJECT MANAGER | **Walk the site together.** Agree where each meter goes. Photograph everything before you start. |
| ☐ | 2 | CUSTOMER | Show the way to the main board and the inverter. Have the router and a socket that stays on ready, and the person who can give the converter a fixed address. |
| ☐ | 3 | ELECTRICIAN | **Apply the customer's lock-out** on the board to be worked on. |
| ☐ | 4 | ELECTRICIAN | **Fit the grid meter** at the supply entry: voltage taps with their small fuses or 2 A breakers as the meter's manual says, and the three current clamps around the incoming phases, **arrow towards the site**. Figure S-5. |
| ☐ | 5 | ELECTRICIAN | **Fit the solar meter** on the inverter's AC output, after the inverter and before the distribution board: the same, clamps with the arrow towards the site. |
| ☐ | 6 | ELECTRICIAN | **Fit the load meter** (only if the owner wants the site's own consumption measured) on the feed to the loads, as its manual says. |
| ☐ | 7 | CUSTOMER | **Battery (if there is one):** the customer's installer, or the battery's maker, enables the battery's Modbus connection and gives you its address, unit number and register table. Nobody from Greecon opens the battery or touches its DC wiring. |
| ☐ | 8 | PROJECT MANAGER | **Mount the enclosure** and in it the converter, the 24 V supply and the UPS (the box too, if there is room, otherwise on a shelf beside it). Figure S-4. |
| ☐ | 9 | PROJECT MANAGER | **Run the RS-485 data cable** from the converter to the grid meter, on to the solar meter, and on to the load meter if there is one: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. The electrician connects its ends at the meters' A and B terminals inside the board. Figure S-2. |
| ☐ | 10 | PROJECT MANAGER | **Set the converter:** Modbus TCP to RTU gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502, and a fixed address on the site's network. Write the address down and photograph the settings page. |
| ☐ | 11 | PROJECT MANAGER | **Connect the network:** plug the box and the converter into the customer's router, or into the sockets of the 4G router if the site has no usable internet. Figure S-3. |
| ☐ | 12 | ELECTRICIAN, PROJECT MANAGER | **Photograph before closing:** every meter's label (model and serial number), the wiring, the converter's settings. Then close the board. |
| ☐ | 13 | ELECTRICIAN | **Remove the lock-out** only when you both agree the work is closed up. |

| | # | Who | Do this |
|---|---|---|---|
| ☐ | 14 | PROJECT MANAGER | **Set each meter at its own display:** unit number (grid 1, solar 2, load 4), speed 9600, parity none. Write them down and photograph each display. Do this before the box starts reading: meters are delivered as unit 1, and two on one cable would answer at once. |
| ☐ | 15 | PROJECT MANAGER | **Power up** the UPS, the 24 V supply, the converter and the box. |
| ☐ | 16 | ELECTRICIAN | **Stay until commissioning is done.** |
| ☐ | 17 | PROJECT MANAGER | **Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list. |

# Addresses and registers

Meters behind one converter share its IP address and differ only by unit number. Write the numbers down here as you set them; you type them into the setup list in step 3.

| Device in Greecon | Make and model | Unit number | IP address | Port |
|---|---|---|---|---|
| **Grid meter** | Eastron SDM630 | 1 | (the converter's) | 502 |
| **Solar meter** | Eastron SDM630 | 2 | (the converter's) | 502 |
| **Load meter (optional)** | Eastron SDM630 | 4 | (the converter's) | 502 |
| **Battery system (optional)** | | (from its manual) | (its own, or the converter's) | |

## The Eastron meters

Their registers are already filled in by the template: total active power, input register 52 (a manual's 30053; counted from zero), a 32-bit floating point number with the high half first, in watts, multiplied by 0.001 to make kW. Check them against the meter's manual only if a reading looks wrong. Test one from the box:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type float32
```

## The battery, if there is one

Two readings, mapped by hand from the maker's register table (step 4 of the setup list): press **Choose the register** under each reading's name. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx; the address is counted from zero.

| Reading | Register kind | Address | Kind of number | How it becomes the reading | Notes |
|---|---|---|---|---|---|
| **Battery charge** | | | | | percent |
| **Battery power** | | | | | kW, positive while charging |

**Remove what the site does not have:** open **Equipment and readings**, find the battery (or the load meter) under **Equipment**, and press **Remove**. A reading with no register holds the setup list at step 4. If the plant is not allowed to send energy to the grid, tick *This site cannot send energy to the grid* in the prices (step 1).

## The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to ten steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself when it sees them work: you press buttons only for what a person must check. A step opens to show its detail and stays as you leave it.

| | Step | Title on the setup list | Done by |
|---|---|---|---|
| ☐ | 1 | **Prices from the customer's bills** | Project manager (optional) |
| ☐ | 2 | **Install the box** | Project manager |
| ☐ | 3 | **Where each device is on the network** | Project manager |
| ☐ | 4 | **Which register holds which reading** | Project manager |
| ☐ | 5 | **Every device answers the box** | The platform, by itself |
| ☐ | 6 | **Readings arrive and look right** | The platform, by itself |
| ☐ | 7 | **Someone is told when the site needs attention** | Project manager |
| ☐ | 8 | **Grid meter direction checked** | Project manager, with the electrician |
| ☐ | 9 | **Alert limits agreed with the customer** | Project manager (only if the plant has a battery) |
| ☐ | 10 | **Commissioned** | Project manager |

Below, each step says what to do, what you should see, and what to do if you do not.

### Step 1 Prices from the customer's bills

PROJECT MANAGER *optional*

| | |
|---|---|
| DO | Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**. |
| YOU SHOULD SEE | The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading). |
| IF YOU DO NOT | It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct. |

## Step 2 Install the box

PROJECT MANAGER

| | |
|---|---|
| DO | Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day. |
| YOU SHOULD SEE | **Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box. |
| IF YOU DO NOT | ▪ *Waiting for the install command to be run on the box.* The line has not been run yet.<br>▪ *The box has its credential but has not reported yet.* Wait a minute and check the network cable.<br>▪ *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`.<br>▪ *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`. |

## Step 3 Where each device is on the network

PROJECT MANAGER

| | |
|---|---|
| DO | Under each device, enter its **IP address** (the converter's address), its **Port** (502 unless the converter's page says otherwise) and its **Unit number** (the number you gave the device, 0 to 247). Devices behind one converter share its IP address and differ only by unit number. Press **Save** under each. The box picks up a change within a minute. |
| YOU SHOULD SEE | **Done**: *Every device has an address,* and each device reads like *192.168.1.50:502, unit 1*. |
| IF YOU DO NOT | *N devices still need an address.* A wrong address is refused in words, for example *The address must be an IP address such as 192.168.1.50, or a host name.* |

## Step 4 Which register holds which reading

PROJECT MANAGER

| | |
|---|---|
| DO | For equipment Greecon knows (the Eastron meters in the template) the registers are already filled in: check them against the meter's manual only if a reading looks wrong. For anything else, press **Choose the register** under the reading's name and fill in: the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero, so 30053 is address 52 and 40001 is address 0), the kind of number (and, for two-register numbers, which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**. |
| YOU SHOULD SEE | **Done**: *Every reading is tied to a register.* |
| IF YOU DO NOT | *N readings not tied to a register yet*, each marked *No register chosen.* A reading the site does not have (a flow meter that was not fitted, a battery that does not exist) holds the list at this step: open the site, find it under Equipment, in Devices or Reading points, and press Delete. |

## Step 5 Every device answers the box

THE PLATFORM

| | |
|---|---|
| DO | Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute. |
| YOU SHOULD SEE | Each device shows **Done**: *Answering (N readings)*. On the box, `greecon-edge status` lists each device as **ok**. |
| IF YOU DO NOT | The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2)*. Look in this order: a wrong unit number (the commonest), a wrong speed or parity at the converter, A and B swapped on the data cable, the converter not in gateway mode, a wrong IP address. Nothing answers at all: the IP address or port is wrong, the converter is off, or it is on another network than the box. Some readings work and others do not: a register address is wrong for that device. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon. |

## Step 6 Readings arrive and look right

THE PLATFORM

| | |
|---|---|
| DO | Nothing to press. Look at each reading: is it a number you believe? |
| YOU SHOULD SEE | Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.* |
| IF YOU DO NOT | *No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *237 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* That is almost always a wrong scale, a wrong register or a wiring fault. |

## Step 7 Someone is told when the site needs attention

PROJECT MANAGER

| | |
|---|---|
| DO | Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Make sure each person on the list also has a login of their own with the Operator role, so that they can press **Acknowledge**. Then, back on the setup list, press **I checked this**. |
| YOU SHOULD SEE | After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent. |

| | |
|---|---|
| IF YOU DO NOT | ■ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned.<br>■ *Nobody is set to be told:* add the people.<br>■ *Nobody on the list has an email address* (or a phone number): add one for each person.<br>■ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam.<br>■ *Test messages are limited to 6 an hour.* Wait and try again. |

## Step 8 Grid meter direction checked

PROJECT MANAGER | ELECTRICIAN

| | |
|---|---|
| DO | Greecon cannot see which way the meter's clamps point. With the sun down or the solar switched off, switch on a known load (a kettle or a heater). **Grid import must go up.** If **Grid export** goes up instead, the clamps are the wrong way round: the electrician turns them (with the lock-out on) and you check again. Then press **I checked this** with a note of what you saw. |
| YOU SHOULD SEE | Before: **To do**, *Greecon cannot see which way the meter's clamps point.* After you confirm: **Done**, *Checked on site,* with your name and the time. |
| IF YOU DO NOT | Import and export swapped: the clamps are the wrong way round. The electrician turns them; you repeat the check. A reversed clamp swaps what the site buys with what it sells, and every figure with it. |

## Step 9 Alert limits agreed with the customer

PROJECT MANAGER *only if the plant has a battery*

| | |
|---|---|
| DO | The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**. |
| YOU SHOULD SEE | Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard).* After you confirm: **Done**, *Checked on site.* |
| IF YOU DO NOT | The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered. |

## Step 10 Commissioned

PROJECT MANAGER

| | |
|---|---|
| DO | Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note. |

| | |
|---|---|
| YOU SHOULD SEE | The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears. |
| IF YOU DO NOT | While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first. |

# Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it, and check again.
- When every line passes, press **Commission this site** on the setup list (step 10) with a note. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

| Check and how | Pass when | Greecon shows | Real | Result |
|---|---|---|---|---|
| **Grid meter's display against Greecon.** Read the display (kW or W) and Greecon at the same time. | Within 5 percent, or 1 kW on a small load | | | ☐ Pass ☐ Fail |
| **Solar meter's display against Greecon.** The same, on the solar meter. | Within 5 percent, or 1 kW on a small load | | | ☐ Pass ☐ Fail |
| **Load meter's display against Greecon** (if fitted). | Within 5 percent, or 1 kW on a small load | | | ☐ Pass ☐ Fail |
| **Grid direction.** A known load on (a kettle or a heater), the sun down or the solar off. On a plant that exports at noon, do it at night or with the inverter off. | Import rises; export does not | | | ☐ Pass ☐ Fail |
| **Solar at night.** | Zero, or a few watts | | | ☐ Pass ☐ Fail |
| **Solar at midday.** Against the inverter's own display. | Within about 5 percent; grid export is the difference between production and what the site is using | | | ☐ Pass ☐ Fail |
| **Battery charge** (if fitted). Against the battery's own display. | The two agree (write both numbers) | | | ☐ Pass ☐ Fail |
| **A test message.** Each person on the list. | It arrived, in the inbox and on the phone | | | ☐ Pass ☐ Fail |
| **Power cut.** Unplug the box for 30 seconds, plug it in. | Back reporting within 3 minutes, and the box says nothing about readings lost | | | ☐ Pass ☐ Fail |
| **Internet cut** (the first install, and any time the box or its software is new). See the box below. | A queue while it is out; empty a few minutes after it returns | | | ☐ Pass ☐ Fail |

DO

**The internet cut, step by step.** Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Signed off by: _____ Date: _____

## Handover

About 30 minutes at the customer's. PROJECT MANAGER with the owner, and the owner's people who are told.

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told and how to reach them, what was commissioned and by whom, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); **Energy now**, the flow moving from the solar to the site and the grid; **Value**, what the solar kept on site this month and the prices it uses; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each person on the list has a login of their own, so that they can acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Say the sentence** below.
- ☐ **Say how alerts reach people,** in the words below.
- ☐ **Leave:** the printed sheet, signed; the box's label; a card with `info@greecon.earth` and your phone.
- ☐ **File:** the photographs and the survey form go in the shared drive, in a folder named for the customer.
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

*Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.*

SAY

*How alerts reach people:* **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Acknowledging means pressing* **Acknowledge** *on the* **Alerts** *page, so each person on the list needs a login of their own (the Operator role is enough).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

**Never promise a saving, a percentage or "AI".** Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

**Their data is theirs.** Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

# Handover record

KIT S · SOLAR PLANT WITH GRID CONNECTION

| Field | Answer |
|---|---|
| **Customer and site** | |
| **Date, and who was there** | |
| **Who is told, in order (name, mobile with country code, email)** | |
| **Prices entered (electricity, export) and where they came from; zero-export box ticked if needed** | |
| **Battery: limits agreed (Low below), or no battery** | |
| **Setup list: every step Done and the site Commissioned** | ☐ Yes |
| **Handover sheet printed and signed** | ☐ Yes |
| **Shown: Overview, Energy now, Value, Who is told with a live test message, Reports** | ☐ Yes |
| **Left: label on the box; card with `info@greecon.earth` and your phone** | ☐ Yes |
| **Photographs and survey form filed in the customer's folder** | ☐ Yes |

**Project manager: name and signature**

**Owner: name and signature**

# The three most likely problems

| What you see | Likely cause | What to do |
|---|---|---|
| **The solar meter reads zero at midday** | A clamp closed on the wrong cable, a closed breaker upstream, or the inverter is limited to zero export | Compare with the meter's own display and test the register from the box ( `greecon-edge modbus-test` ). Ask the electrician to confirm that the clamps are on the inverter's output phases. If the plant is zero-export, midday solar can honestly be low: check the inverter's display |
| **Import and export are swapped** (export rises when a load is switched on) | The grid meter's clamps are the wrong way round | The electrician turns them, with the lock-out on; you repeat the direction check (step 8 of the setup list). A reversed clamp swaps what the site buys with what it sells, and every figure with it |
| **A reading is a thousand times too big** | Watts shown as kilowatts, typically on a hand-mapped battery reading | In the register form multiply by 0.001 (the Eastron meters already do) |

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: a wrong unit number (the commonest), a wrong speed or parity at the converter, A and B swapped on the data cable, the converter not in gateway mode, a wrong IP address.